Population PK-PD Modeling & Simulation of Metoprolol: Mixture Models, Sub-Populations and Phenotype Classifications

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Outline

1. Motivating Example
2. Simulation study of mixture models and classification
3. Coding mixture models

Objective# 1

To develop Population PK and Population PK-PD models for metoprolol and to specifically quantitate the effects of CYP2D6 phenotype and smoking.

Introduction

- Metoprolol, a cardio-selective beta-blocker
  - ~ 70% metabolized by CYP2D6†
  - ~ 6-8% North American Caucasians poor metabolisers of CYP2D6
- Smokers have improved outcomes relative to nonsmokers ‡
  - PK or PD effect?
- Two studies conducted
  - Characterization of PK & PD of metoprolol
  - CYP2D6 phenotype & smoking status

†Johnson JA, Burlew BS, Metoprolol Metabolism via Cytochrome P4502D6 in Ethnic Populations, Drug Metabolism and Disposition, 24 pg.350-355, (1996).

‡John Wikstrand, Ingrid Warnold, Gunnar Olsson etal., 'Primary Prevention with Metoprolol in Patients with Hypertension', JAMA, 259(13), pg. 1976-1982, (1988).

Study Design

- 243 healthy males screened
- CYP2D6 Phenotype status (EM / PM)
  - Dextromethorphan as probe
- Smoking status (SM / NS)
  - At least 1 pack of cigarette a day at the time of recruitment

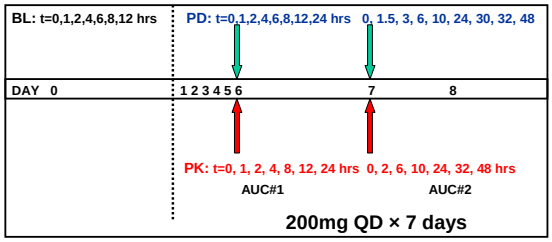
Demographics	Mean	Range
Weight (kg)	80	59 – 104
Height (cm)	179	164 – 191
Age (yr)	27	22 – 42
Covariates	EM	PM
SM	7	6
NS	10	12

EM – Extensive metaboliser, PM – Poor metaboliser, SM – Smokers, NS - Nonsmokers

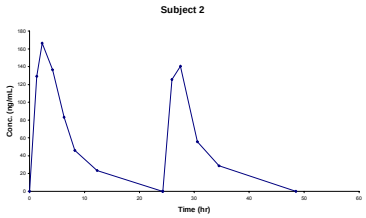
Data & Study Design

- All subjects received 200mg metoprolol orally QD for 7 days
- Day 0 (baseline)
  - 7 sequential baseline (no drug) heart rate measurements
  - Exercise protocol (6 mph, 2.5 min, 10 degree incline motorized treadmill test)
- Day 6 & 7
  - 13 S(-)-metoprolol plasma concentrations
  - 16 Exercise-Induced Heart Rates (EIHR)

Concentration-Effect Sampling Scheme



Typical Data Set



Population PK Modeling

- 1-CMT, 1st-order oral absorption & elimination model
- NONMEM version V, level 1.1, S-PLUS 2000
- Model building guided by:
  - Statistically significant reduction in objective function value (α=0.01) with FOCE-I
  - Visual inspection of the diagnostic plots

Population PD Modeling

- PD marker is the reduction in Exercise-Induced Heart Rate (EIHR)
- Sequential approach (2 Steps)
  - Individual PK parameters estimated & fixed
  - Estimate Population PD parameters
- Direct and Link PD models

Population PK Parameter Estimates

PARAMETER	ESTIMATES (%RSE)
CL-PM (L/hr)	98 (7%)
CL-EM (L/hr)	267 (7%)
V (L)	942 (5%)
KA (hr^{-1})	3.0 (17%)
ALAG (hr)	0.8 (3%)
SMK(fraction)	0.80 (6%)

EM – Extensive metaboliser, PM – Poor metabolizer, SMK – Smoking

Conclusions

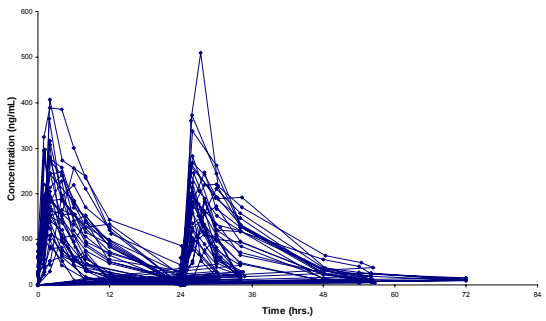
- CYP2D6 poor metabolizers had 60% lower metoprolol clearance compared to extensive metabolizers.
- Smokers on average had 20% lower clearance irrespective of the phenotype status.
- CYP2D6 phenotype and smoking status do not significantly affect the pharmacodynamic parameters of metoprolol.
- Concentration-effect relationship for metoprolol was best described by a direct baseline inhibitory Emax model.

Gosh, Dr. Brundage, phenotype made a huge OFV difference. We don't usually have the PT. What then?

Well, Nitin, let me tell you about mixture models.

Mixture Models

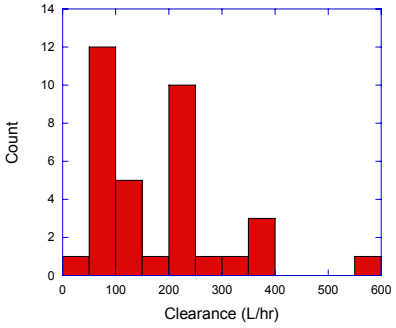
- Mixture models in NONMEM help in identifying sub-populations when there is no available covariate that distinguishes between the sub-populations.
- Nature of metoprolol Pop PK dataset led to the exploration whether a mixture model could detect subpopulations in the distribution of clearance.

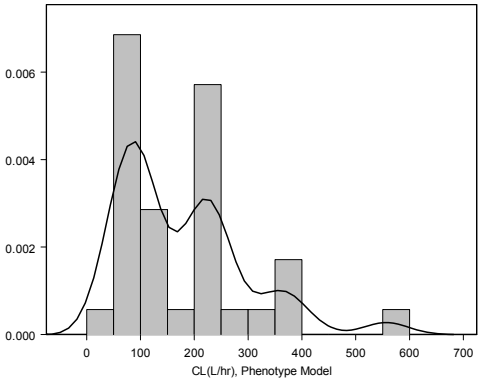


Mixture Models

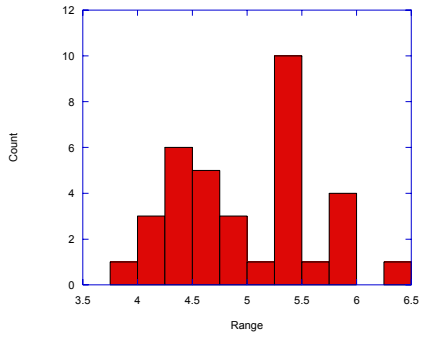
- With mixture models, it is important to remember that there doesn't need to be a physiological basis for the subpopulations
- It can simply reflect a broad distribution, with no true subpopulations

Distribution of Clearances

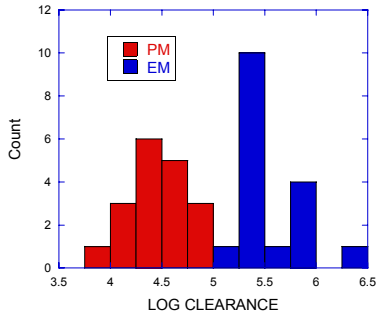




Distribution of Log CL



CL Distribution by Phenotype



Phenotype Model vs. Mixture Model

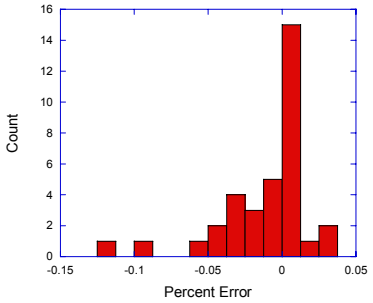
PHENOTYPE POP-PK MODEL	METOPROLOL MIXTURE MODEL
CL EM/NS = 267 L/hr	CL POP2/NS = 267 L/hr
CL EM/SM = 214 L/hr	CL POP2/SM = 214 L/hr
CL PM/NS = 98 L/hr	CL POP1/NS = 98 L/hr
CL PM/SM = 78 L/hr	CL POP1/SM = 78 L/hr
V = 942 L	V = 943 L
KA = 3.0 hr-1	KA = 3.0 hr-1
ALAG = 0.80 hr	ALAG = 0.80 hr
PM/(PM+EM) = 18/35 = 0.514	POP1 Mixing Fraction = 0.515

*Kaila N., Brundage R.C., Straka R. J., 'Population Pharmacokinetic Modeling of Metoprolol: Mixture models and CYP2D6 Phenotype', AAPS abstract, 2003.

Individual post-hoc CL estimates

ID	CL_phenotype model	CL_mixture model	Phenotype Status
7	44.97	44.96	PM
31	65.47	65.49	PM
30	67.10	67.09	PM
24	68.37	68.35	PM
23	71.73	71.74	PM
34	74.06	74.06	PM
21	74.35	74.36	PM
9	76.60	76.59	PM
20	78.33	78.32	PM
32	78.49	78.48	PM
35	91.03	91.03	PM
11	96.02	96.05	PM
10	99.63	99.63	PM
8	102.58	102.58	PM
25	107.53	107.55	PM
22	116.52	116.55	PM
12	117.51	117.56	PM
33	119.47	119.50	PM
17	189.08	189.03	EM
18	208.89	208.90	EM
2	213.32	213.30	EM
13	215.84	215.83	EM
15	220.18	220.20	EM
4	222.45	222.43	EM
19	224.50	224.50	EM
16	229.29	229.29	EM
27	230.40	230.43	EM
6	236.35	236.33	EM
3	242.15	242.20	EM
1	254.42	254.42	EM
26	343.88	343.94	EM
29	358.40	358.63	EM
28	361.12	361.27	EM
14	395.76	396.11	EM
5	556.47	557.15	EM

Distribution of % Deviations in CL



OK, Dr. Brundage, the CLs match great, but how do I know if someone is in Pop1 or Pop2?

Well, Nitin, let me tell you about group assignments in mixture models.

Sub-Population Assignments

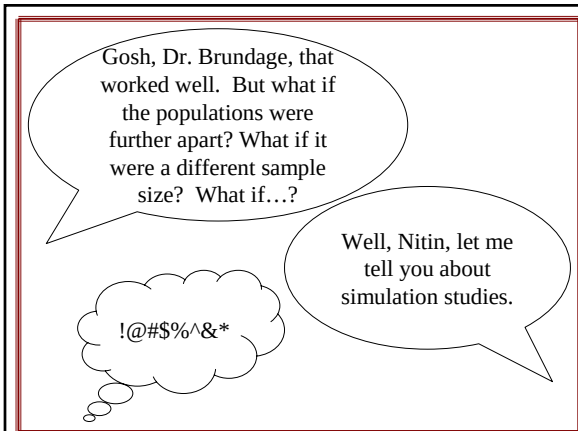
- NONMEM will calculate the likelihood of each subject's data for each sub-population
- It will assign each subject to the most probable sub-population
- All PRED, RES, WRES, etc. will be computed based on the typical values for that sub-population
- The population number can be tabled

Individual post-hoc clearance estimates and subpopulation assignment

ID	CL_phenotype model	CL_mixture model	Phenotype Status	Subpopulation
7	44.97	44.96	PM	1
31	65.47	65.49	PM	1
30	67.10	67.09	PM	1
24	68.37	68.35	PM	1
23	71.73	71.74	PM	1
34	74.06	74.06	PM	1
21	74.35	74.36	PM	1
9	76.60	76.59	PM	1
20	78.33	78.32	PM	1
32	78.49	78.48	PM	1
35	91.03	91.03	PM	1
11	96.02	96.05	PM	1
10	99.63	99.63	PM	1
8	102.58	102.58	PM	1
25	107.53	107.55	PM	1
22	116.52	116.55	PM	1
12	117.51	117.56	PM	1
33	119.47	119.50	PM	1
17	189.08	189.03	EM	2
18	208.89	208.90	EM	2
2	213.32	213.30	EM	2
13	215.84	215.83	EM	2
15	220.18	220.20	EM	2
4	222.45	222.43	EM	2
19	224.50	224.50	EM	2
16	229.29	229.29	EM	2
27	230.40	230.43	EM	2
6	236.35	236.33	EM	2
3	242.15	242.20	EM	2
1	254.42	254.42	EM	2
26	343.88	343.94	EM	2
29	358.40	358.63	EM	2
28	361.12	361.27	EM	2
14	395.76	396.11	EM	2
5	556.47	557.15	EM	2

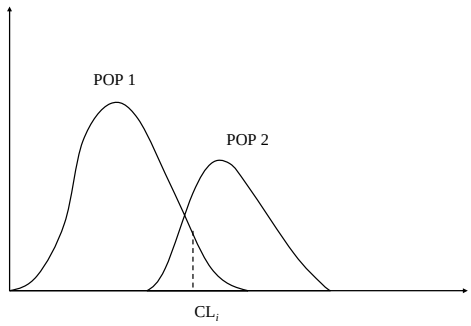
All PM classified into population 1

All EM classified into population 2



OBJECTIVE# 3

To conduct a **simulation study** to evaluate subpopulation assignments when using a mixture model in NONMEM



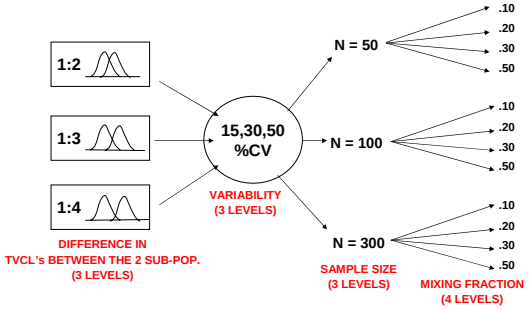
Objective

- To determine the effect of 4 factors on the correct classification of subjects by a mixture model in NONMEM under a sparsely sampled design:
  - Difference in the mean clearance
  - Mixing fraction of population
  - Variability in clearance
  - Population sample size

Methodology

- **Simulate concentration-time data for two subpopulations (PM:EM), ratio of their mean clearance (1:2; 1:3; 1:4). (3 levels)**
- **Proportion of subjects in 1st subpopulation versus 2nd was 0.1, 0.2, 0.3 and 0.5. (4 levels)**
- **IIV in clearance modeled exponentially as 15%, 30% and 50% coefficient of variation (CV). (3 levels)**
- **Sample sizes (N) of population**
  - **50, 100 and 300 subjects (3 levels)**

Flow Diagram

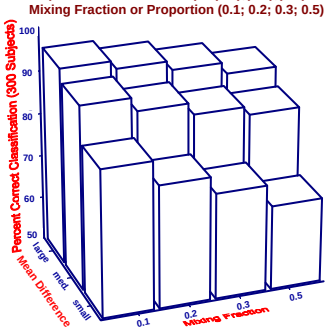


Methodology

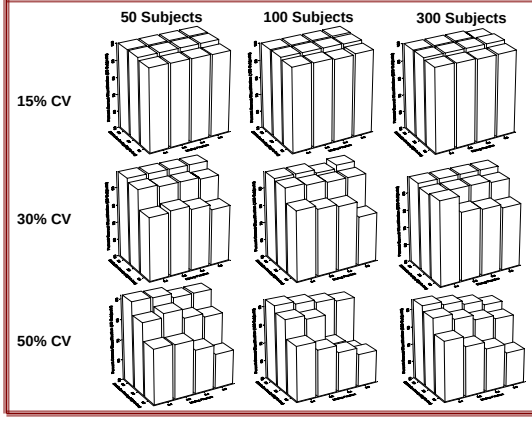
- PopPK model and sampling for simulation
  - Single dose, 1-CMT, 1st order absorption and elim
  - Sampling at t = 2,8,12,24 hours (± 0.25 SD)
- Residual variability modeled using a combined error model (10% CV, 0.01 ng/mL)
- 300 datasets simulated ($SIM) at each of the 3×4×3×3 = 108 combinations
- 300 datasets for each combination were estimated by a mixture model on clearance
- Percent correct classification for each dataset at each combination was calculated

Percent Correct Classification

(N=300;CV=50%)
(TVCLPOP1:TVCLPOP2) = (1:2);(1:3);(1:4)
Mixing Fraction or Proportion (0.1; 0.2; 0.3; 0.5)



-% CC = (Subject correctly classified in successfully converged runs with covariance matrix / Total number of subjects in those runs) * 100 %.



Conclusions

- As the difference in mean clearances increases, subjects are more likely to be correctly classified.
- Variability in the subpopulations and correct classification are inversely related.
- Larger sample size is more likely to have correct classification.
- As the proportion of subjects in the subpopulation approaches 0.5, more subjects tend to be misclassified.

Coding Mixture Models

```
$PK
 EST=MIXEST
 IF(MIXNUM.EQ.2) THEN
   TVCL=THETA(1)
   CL=TVCL*EXP(ETA(1))
   TVV=THETA(2)
  ELSE
   TVCL=THETA(3)
   CL=TVCL*EXP(ETA(2))
   TVV=THETA(4)
 ENDIF
 V=TVV*EXP(ETA(3))
```

Coding Mixture Models

```
$MIX
 NSPOP = 2
 P(1)  = THETA(5)
 P(2)  = 1-THETA(5)

$TABLE  … EST …
```